Fine-tuning dataset: incident-report classifier

Form pack: Qualcomm HQ incident reporting (`incident_taxonomy.json` , 28 incident types). Dataset version 1.0, October 2026.

1. What it is

A set of **multi-turn, hand-written English conversations** between an employee reporting a facility problem and an assistant that asks follow-up questions. It trains the on-device classifier (recommended model: Ettin-encoder-17m, trimmed vocab; see `bench/RESULTS.md`) to do three things after every user turn:

Task	Label	Where it comes from
Incident type	one of 28 type ids, `undetermined` (not clear yet) or `none` (not an incident)	`type_known` on each user turn
Sentence relevance	relevant / filler for each user sentence	`relevant` on each sentence
Form fields mentioned	multi-label: which form fields a sentence gives information about	`fields` on each sentence

Filling in the field *values* is not the classifier's job (an LLM does that from the relevant sentences), so values are not labelled.

Size and composition

Part	Count	Notes
28 types × 32 `single`	896	normal reports
28 types × 6 `confusable`	168	start out sounding like a neighbouring type, then resolve
`none`	60	not an incident: wrong app, chit-chat, IT help, mic tests
`multi`	40	two incidents in one report (primary + secondary type)
Train total	**1,164**	
Test (4 per type + 10 none + 6 multi)	128	unseen reporters, places and situations
Total	**1,292 conversations**	2,976 user turns (one training sample each), 5,544 labelled sentences

How a training sample is built

One sample **per user turn**: the input is *all user turns so far* joined in order (assistant turns are **not** included, see K2). Targets: that turn's `type_known` ; relevance and fields for every sentence so far (broadcast to the sentence's tokens). Split by **conversation**, never by turn, so no conversation is in both train and validation.

How it was made

- Two LLM authors wrote every conversation by hand, sentence by sentence, with no templates or generation scripts: Gemini (Antigravity) started five types (water_leak, washroom_plumbing, power_outage, electrical_hazard, lighting_fault) and Claude wrote everything else. In those five types, half of the conversations were rewritten by Claude with invented locations, so **every type contains both writing styles and both location kinds** (conditions L1, K3, S2). `datagen/compile.py` only converts Claude's plain-text format to JSONL.

- Every file passes `datagen/validate_gen.py` (format, label validity, no repeated sentence of 5+ words anywhere in the dataset) and is audited by `datagen/audit.py` (the balance conditions in section 3).
- A label review found that vague symptom openings ("the wall plate was scorching hot") had been marked as filler. 56 sentences were relabelled to follow rule R4; the pre-review copy is kept in `datagen/out.bak-gemini/`.

2. The case: Qualcomm HQ form pack

What we assumed

The first form pack models an **incident reporting assistant for a Qualcomm campus**. Everything in it is fixed configuration in `incident_taxonomy.json`; the AI never invents types, fields or routes.

- **Organisation:** departments modelled on Qualcomm's Facilities & Security Group (Facilities; Environment, Health & Safety; Security), with IT and R&D Labs as partner orgs.
- **Campus:** buildings identified by letter codes, **N** (floors 1–12) and **QRC** (floors 1–6); 14 space types (conference room, workstation bay, R&D lab, server/IDF room, pantry, cafeteria, washroom, lobby/reception, elevator lobby, stairwell, corridor, parking, loading dock, outdoor).
- **Demo values:** priorities, SLAs, floors, space types and technician rosters were chosen by the team. They are **not** Qualcomm internal data.
- **Division of work:** the **classifier** picks the incident type and the relevant sentences; an **LLM** fills only the fields listed for that type, quoting the transcript span each value came from; **code** applies the priority and routing rules.
- **Conversation policy:** at most 4 follow-up questions, at most 2 fields per question, asked in the order safety-critical → location → type-specific → time. "I don't know" is allowed; an unknown safety field is treated as the worse case.

Taxonomy structure

Section	Contents
`meta`	name, version, org model, design rule, demo-values note
`priorities`	P1–P4 with response SLA and meaning
`completion_policy`	follow-up limits and question order (above)
`location_model`	buildings, floors, space types; required: building, floor, space_id
`common_fields`	fields every report has: building, floor, space_type, space_id, landmark, occurred_at, immediate_actions_taken, persons_involved, witnesses, asset_tag, plus system fields (incident_type, description, reporter)
`field_library`	~50 type-specific fields: type (enum / bool / string / int / time), allowed values, safety flag, question text
`incident_types`	the 28 types below: id, label, department, role, default priority, required/optional fields, rules, seed phrases
`global_rules`	unknown safety field → worse case; injury or fire words → add secondary incident; classifier confidence < 0.6 → ask the user to confirm between top-2 types
`departments`, `assignment`	who gets the ticket, when it is due, where it is on the map

Priorities

	Label	Response SLA	Meaning
P1	Critical	15 min	Risk to life, active hazard, or site-wide outage. Page on-call immediately.
P2	High	60 min	Spreading damage, security exposure, or work stopped for many people.
P3	Medium	240 min	Contained problem affecting a room or small group.
P4	Low	1440 min	Cosmetic or convenience issue. Next business day.

Departments

ID	Department	Roles
FAC	Facilities Engineering	Plumbing Technician, Electrical Technician, HVAC Technician, Civil & Interiors Technician, Elevator Service (vendor)
HWS	Housekeeping & Workplace Services	Housekeeping Staff, Cafeteria & Pantry Operations, Pest Control (vendor)
SEC	Global Security	Security Officer, Security Systems Technician (access control, CCTV), GSOC Analyst (Global Security Operations Center)
EHS	Environment, Health & Safety	EHS Engineer, Emergency Response Team / First Aid
ITW	IT & Workplace Technology	IT Support Technician, Conference Room AV Technician, Network Engineer
LAB	R&D Lab Operations	Lab Technician, Lab Facilities Engineer (bench power, UPS, chillers)

The 28 defined incident types

`*` = safety-critical field (asked first). "Confusable with" = the groups the dataset deliberately writes look-alike conversations for. Rules in the taxonomy can raise the priority (for example a water leak near electrics becomes P1, a snake sighting becomes P1).

#	ID	Label	Dept	Default	Required fields	Optional	Confusable with
1	water_leak	Water leak	FAC	P3	`leak_source`, `water_spread`, *`near_electrical`*	`immediate_actions_taken`	washroom_plumbing
2	washroom_plumbing	Washroom plumbing fault	FAC	P3	`fixture_type`, `plumbing_issue`	–	water_leak
3	power_outage	Power outage	FAC	P3	`outage_scope`, `critical_equipment_affected` *	–	electrical_hazard, lighting_fault
4	electrical_hazard	Electrical hazard	FAC	P1	`hazard_kind`, `smoke_present` *	`immediate_actions_taken`	fire_smoke, lighting_fault, power_outage
5	lighting_fault	Lighting fault	FAC	P4	`lighting_scope`	–	electrical_hazard, power_outage
6	hvac_issue	Temperature / HVAC issue	FAC	P3	`hvac_condition`, `affected_scope`	–	–
7	elevator_fault	Elevator fault	FAC	P2	`elevator_id`, `persons_trapped` *	–	–
8	structural_damage	Building damage	FAC	P3	`damage_kind`, `falling_or_injury_risk` *	–	unsafe_condition

#	ID	Label	Dept	Default	Required fields	Optional	Confusable with
9	furniture_fixture	Furniture / fixture repair	FAC	P4	`furniture_item`	–	–
10	spill_cleanup	Spill (non-hazardous)	HWS	P3	`spill_substance`, *`slip_hazard`*	–	chemical_spill
11	washroom_hygiene	Washroom hygiene / supplies	HWS	P4	`hygiene_issue`	–	cafeteria_issue, pest_sighting
12	pest_sighting	Pest sighting	HWS	P3	`pest_kind`	–	cafeteria_issue, washroom_hygiene
13	cafeteria_issue	Cafeteria / pantry issue	HWS	P3	`cafeteria_issue`, `food_safety_concern` *	–	pest_sighting, washroom_hygiene
14	access_control_fault	Door / badge reader fault	SEC	P3	`door_state`, `secure_area` *	–	cctv_fault, lost_badge
15	lost_badge	Lost badge	SEC	P3	`last_seen`	–	access_control_fault
16	suspicious_activity	Suspicious person / tailgating	SEC	P2	`person_description`, `behavior` *	`witnesses`	theft_missing_asset
17	theft_missing_asset	Theft / missing asset	SEC	P3	`asset_kind`, `contains_confidential` *	`asset_tag`	suspicious_activity
18	cctv_fault	CCTV camera fault	SEC	P3	`camera_location`	–	access_control_fault
19	injury_medical	Injury / medical emergency	EHS	P2	`injury_kind`, *`person_conscious`*, `persons_count`	`persons_involved`, `witnesses`, `immediate_actions_taken`	unsafe_condition
20	fire_smoke	Fire / smoke	EHS	P1	`fire_visible` *, `evacuation_status`	`alarm_triggered`	electrical_hazard
21	chemical_spill	Chemical / hazardous spill	EHS	P2	`substance_name`, *`spill_quantity`*, *`persons_exposed`*	–	spill_cleanup
22	unsafe_condition	Near miss / unsafe condition	EHS	P3	`hazard_description`	`witnesses`	injury_medical, structural_damage
23	network_outage	Network outage	ITW	P3	`network_scope`, `connection_type`	–	conference_room_av, device_fault
24	conference_room_av	Conference room AV fault	ITW	P4	`av_equipment`, `meeting_in_progress`	–	device_fault, network_outage
25	device_fault	Printer / shared device fault	ITW	P4	`device_kind`	`asset_tag`	conference_room_av, network_outage

#	ID	Label	Dept	Default	Required fields	Optional	Confusable with
26	lab_equipment_fault	Lab equipment fault	LAB	P3	`equipment_kind`, `test_in_progress`	`asset_tag`	esd_issue, lab_utility_failure
27	esd_issue	ESD control issue	LAB	P3	`esd_issue`	–	lab_equipment_fault, lab_utility_failure
28	lab_utility_failure	Lab utility failure	LAB	P2	`lab_utility`, `test_in_progress`	–	esd_issue, lab_equipment_fault

Two extra classifier labels exist outside the taxonomy: **`undetermined`** (the user hasn't said enough yet) and **`none`** (not a facility incident at all).

3. Conditions the data must meet (anti-overfitting, anti-imbalance)

Checked automatically by `python3 datagen/audit.py` unless marked *manual*.

Class balance

- **C1** Same number of training conversations per type (max/min ≤ 1.2). Target 38 each.
- **C2** Enough out-of-scope (`none` ≥ 60) and `multi` (≥ 40) examples, so the model can say "not an incident".
- **C3** ≥ 6 confusable conversations per type, covering the 12 confusable groups in both directions.
- **C4** *(manual)* `undetermined` is ~25–30% of first turns; the model must learn to wait, not guess.

Relevance balance

- **R1** Filler is 20–35% of all sentences.
- **R2** Each type's filler rate is within 12–40%: if one type had far more filler, the filler rate would hint at the type.
- **R3** *(manual)* Filler sometimes mentions words from **other** types ("the cafeteria queue was insane" inside an HVAC report), so the model can't classify by keyword alone.
- **R4** *(manual)* One relevance rule everywhere: symptoms, places, times, actions and "I don't know" answers are relevant; feelings, greetings, small talk and wrong guesses are filler.

Structure must not give the type away

- **S1** Mean user turns per type within ±0.6 of the overall mean.
- **S2** Mean words per sentence per type within ±25% of the overall mean.
- **S3** First-turn `undetermined` rate per type between 15% and 40%.
- **S4** Turn mix close to 15% one-turn / 35% two / 30% three / 20% four-plus.
- **S5** *(manual)* Opening styles rotate per type: only *what*, only *where*, greeting, vague symptom, panicked burst, nearly complete (≤ 1 in 6).

Location, people and wording must not give the type away

- **L1** Qualcomm buildings (N, QRC) in 30–75% of each type's conversations; the rest use invented places.
- **L2** *(manual)* Reporters (intern, engineer, guard, housekeeping, lab tech, receptionist, visitor, manager, night shift, contractor) and tones rotate across every type.
- **K1** Filler sentences alone must not predict the type (naive Bayes accuracy ≤ 3× chance).
- **K2** The assistant's questions *do* reveal the type (about 70% accuracy from assistant text alone), so **assistant text must never be model input**.

- **K3** Writing style must not reveal the type. With two authors, each type should contain both styles.
- **D1** No repeated sentence of 5+ words anywhere, and no near-duplicate first turns (`validate_gen.py`).

Field labels

- **F1** Every required field of a type labelled ≥ 10 times in that type.
- **F2** *(manual)* Across each type, every enum value, both values of each yes/no field, and "don't know" answers appear.

Train / test separation

- **T1** No test sentence (4+ words) appears in train.
- **T2** Test set has ≥ 4 conversations per type, with reporters, places and situations not used in training.
- **T3** *(manual)* Split by conversation; calibration (temperature scaling) uses a validation split carved from train, never the test set.

Acceptance

The dataset is accepted for fine-tuning only when `validate_gen.py` reports **0 errors** and `audit.py` reports **0 FAIL**. Warnings must be reviewed and either fixed or explained.

Check	Required result	Result (v1.0)
Format and labels (`validate_gen.py`)	0 errors	**0 errors, 0 warnings**
C1–C3	28 × 38 train, 6 confusables each, `none` 60, `multi` 40	**pass**: 38 per type, 6 confusables each, 60 / 40
R1, R2	filler 20–35% overall, 12–40% per type	**pass**: 23% overall, 15–34% per type
S1, S2, S3	per-type turns, sentence length, undetermined openings within bounds	**pass**: 2.0–2.6 turns, sentence length within ±25% of 7.7 words, 16–29% undetermined openings
S4	≥ 10% one-turn, ≥ 12% four-plus-turn	**pass**: 15% / 12%
L1	N/QRC in 30–75% of each type	**pass**
K1	filler-only naive Bayes ≤ 3× chance	**pass**: 7% vs 4% chance
K3	both styles in every Gemini-started type	**pass**: 50% / 50% in all five
F1	required fields ≥ 10 labels per type	**pass**
T1, T2	128 test conversations, no shared sentence with train	**pass**: 128, 0 shared

Audit summary: 18 PASS, 0 WARN, 0 FAIL. User turns per conversation: 1 → 179, 2 → 702, 3 → 277, 4 → 108, 5 → 26.

4. Assumptions

1. **Input is clean, complete English text** from speech recognition: no Hindi/Hinglish, no ASR errors.
2. **The classifier sees user turns only.** The assistant's questions follow the taxonomy's completion policy, but they are not model input.
3. **Synthetic conversations stand in for real reports.** LLM-written dialogue approximates how employees talk; it has not been compared against real reports.

4. **Balanced classes.** Every type gets the same amount of data. Real incident frequencies differ (lighting faults are far more common than snakes), so deployment may need prior adjustment.
5. **One primary type per conversation.** `multi` conversations have a secondary type, but the type head predicts one label; secondaries are handled by the taxonomy's global rules.
6. `type_known` **is a judgement**: "what a careful human could name with confidence from the user's words so far". Different annotators would sometimes disagree on the turn where it switches.
7. **Fields are mentions, not values.** A sentence is tagged with a field if it says anything about it, including "I don't know".
8. **The taxonomy is fixed** (28 types, field library, demo location model). Changing it means relabelling.

5. Limitations

- **Small.** 38 conversations (~95 turn samples) per type. Enough to fine-tune a pretrained encoder, not to train one from scratch (one reason the LNN was rejected).
- **Two LLM authors.** Mixing both styles inside every type removes style as a shortcut for the type, but both authors are LLMs: real employees will phrase things in ways neither wrote.
- **No independent label check.** The writer labels its own sentences; there is no second annotator and no agreement score. One systematic error (symptoms marked as filler) was found and fixed during review; others may remain.
- **Test set shares the authors.** Accuracy on it will be optimistic. A small **human-written test set** (team members, real phrasing, ideally spoken then transcribed) is needed before trusting any number.
- **No speech noise, accents or code-switching**, although requirement I1 mentions Hinglish.
- **Labels are coarse for relevance.** Relevance is per sentence; long run-on sentences mixing filler and facts are labelled relevant as a whole.
- **Old data excluded.** `data/*.jsonl` (198 templated conversations, 11 types, partly Hindi) is not used for training; it was only used as benchmark input.
- **Pack-specific.** Everything here applies to the Qualcomm HQ pack. A new form pack needs its own dataset that meets the same conditions.

6. Files

Path	What
`datagen/out/train/<type>.jsonl` , `datagen/out/test/`	the dataset
`datagen/src/`	Claude's hand-written source text
`datagen/PLAN.md` , `datagen/PROGRESS.md`	per-conversation plan and progress log
`datagen/validate_gen.py`	format and label checker
`datagen/audit.py`	balance and shortcut audit (section 3)
`datagen/PROMPT_ANTIGRAVITY.md`	the writing rules both authors follow